



Modelling Financial Markets

Seminar 15: LLMs and Sentiment Analysis — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BERT: Bidirectional Encoder Representations from Transformers
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAR: Cumulative Abnormal Return
- ▣ CI: Confidence Interval
- ▣ DELTA: Deep Learning for Text Analytics (Quantinar course)
- ▣ ECE: Expected Calibration Error
- ▣ FNSPID: Financial News and Stock Price Integration Dataset
- ▣ GI: (Harvard) General Inquirer, dictionary IV-4
- ▣ GPT-4: Generative Pre-trained Transformer 4
- ▣ IV-4: Harvard IV-4 (version 4 of the General Inquirer word lists)
- ▣ LDA: Latent Dirichlet Allocation
- ▣ LLM: Large Language Model
- ▣ LM: Loughran-McDonald (financial sentiment dictionary)
- ▣ NLL: Negative Log-Likelihood
- ▣ OLS: Ordinary Least Squares
- ▣ PPI: Prediction-Powered Inference
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ SD: Standard Deviation
- ▣ SE: Standard Error
- ▣ TF-IDF: Term Frequency - Inverse Document Frequency
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

Outline

Introduction

Part A: Derivations with Numbers

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Score a text with a sentiment dictionary and with the label probabilities of a large language model (LLM)
- ▣ Derive the attenuation of a regression slope on a misclassified sentiment label, and the variance and power of paired classifier tests
- ▣ Compare two classifiers with a paired test and detect training-data leakage and look-ahead bias
- ▣ Make inference valid with model scores: prediction-powered inference, wild cluster bootstrap with few clusters, multiplicity and specification curves

- ▣ Background: Lecture 15; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations with numbers (A1–A8)
- ▣ **Part B:** eleven computational exercises with inference (B1–B11)
- ▣ **Part C:** two open questions, seeds for the team project (C1, and C3, a replication of [Lopez-Lira & Tang \(2026\)](#)), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- You may use an AI assistant, but declare it in [AI_USE.md](#) and log the errors you catch in [AI_ERRORS.md](#)
 - ▶ C2 trains exactly this: find the errors in an AI answer about sentiment backtests
- Always ask: was the text public before the return I try to predict, and did the model already know the outcome?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (attenuation from misclassification), A3 (variance and power of a paired test), A5 (probabilities and the optimal temperature), A7 (break-even cost with SE)
- Part B: B1 (FinBERT vs LM, McNemar), B3 (training-data leakage), B5 (clustered standard errors), B7 (does the LLM remember the market?), B9 (prediction-powered inference)

Proposed: you solve them

- Part A: A2 (LM attenuation; model: A1), A4 (power, FinBERT vs Qwen; model: A3), A6 (Fleiss' κ , Dawid–Skene; model: A1), A8 (Qwen after costs; model: A7)
- Part B: B2 (model: B1), B4 (model: B3), B6 (model: B5), B8 (model: B1), B10 (wild cluster bootstrap; model: B5), B11 (specification curve; model: B7)
- Part C: C1, does headline sentiment pay after costs?; C2, critique an AI answer on an LLM sentiment backtest; C3, replicating [Lopez-Lira & Tang \(2026\)](#) on FNSPID headlines (model: C1)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter15_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`; `transformers` only to recompute the scores
- Labelled texts: Financial PhraseBank (4 846 news sentences, 2 264 with full agreement) [Malo et al. \(2014\)](#); Twitter Financial News, validation set (2 388 headlines)
- Dated headlines: FNSPID (Financial News and Stock Price Integration Dataset) [Dong, Fan & Peng \(2024\)](#), 106 820 headlines on 16 large US stocks; daily closes of the stocks and of SPY (an S&P 500 index fund)
- The scores used in Parts A, B, C1 and C2 are precomputed in the tables of Chapter 15 (C3 needs the original FNSPID headlines and a new scoring run); Quantlets: [MFM_ch15_sem_partA](#), [partB](#), [partC](#)

Formulas Used Today (1/2): Scores and Metrics

- Dictionary tone: $\tau = (P - N)/(P + N)$, P , N : counts of positive and negative words; $\tau = 0$ if $P + N = 0$
 - ▶ LM: the Loughran–McDonald financial word lists [Loughran & McDonald \(2011\)](#); GI: the Harvard General Inquirer IV-4 list, a generic psychology dictionary
- LLM label probabilities: $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ (z_k : logit of answer word k , T : temperature); net score $s = p_{\text{pos}} - p_{\text{neg}}$
 - ▶ FinBERT: a BERT (Bidirectional Encoder Representations from Transformers) model fine-tuned for financial sentiment [Araci \(2019\)](#); Qwen2.5: open-weight LLMs [Qwen Team \(2024\)](#)
- Misclassified binary regressor [Aigner \(1973\)](#): $\text{plim } \hat{b} = \beta \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)/[p(1 - p)]$
 - ▶ Fleiss' $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$ for agreement among many raters [Fleiss \(1971\)](#)

Formulas Used Today (2/2): Inference and Signals

- McNemar test [McNemar \(1947\)](#): n_{01} texts where A is wrong and B right, n_{10} the reverse; under H_0 , $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$
 - Bootstrap CI (confidence interval): resample texts 2 000 times, take the 2.5% and 97.5% quantiles
- AUC (area under the ROC curve): probability that a random “up” case gets a higher score than a random “down” case; 0.5 = no information
- PPI [Angelopoulos et al. \(2023\)](#), n labelled and N unlabelled texts sampled independently, f fixed:

$$\hat{\theta} = \frac{1}{N} \sum_{\text{unlab.}} f(\tilde{X}_i) - \frac{1}{n} \sum_{\text{lab.}} (f(X_j) - Y_j), \text{ variance } \widehat{\text{Var}}(f)/N + \widehat{\text{Var}}(f - Y)/n$$
- Panel regression $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$: r^e excess return over SPY (%), z standardised daily score, $h = 0, 1, 2$ days after the news day d
 - Standard errors (SE): ordinary least squares (OLS) or clustered by day or by stock [Petersen \(2009\)](#); bp: basis point, 0.01%
- Strategy: Newey–West t [Newey & West \(1987\)](#); annualised Sharpe ratio $\sqrt{252} \bar{r}/s_r$; break-even cost = mean gross return on days with positions / 4 (four trades a day)

A1: Attenuation from Misclassification (FinBERT) (task 1/2) [Solved]

Task

□ Context and assumptions

- Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- Regression of interest: $r = a + \beta s^* + \varepsilon$, where $s^* = 1$ if a headline is truly negative and 0 otherwise; we only observe $\hat{s} = 1$ when FinBERT predicts “negative”
- Assumption (non-differential error): once s^* is known, $\hat{s}$ carries no information about r , so $\text{Cov}(\varepsilon, \hat{s}) = 0$
- Notation: $\pi = P(s^* = 1)$, $p = P(\hat{s} = 1)$, false-alarm rate $\alpha_0 = P(\hat{s} = 1 \mid s^* = 0)$, miss rate $\alpha_1 = P(\hat{s} = 0 \mid s^* = 1)$; FinBERT confusion matrix (rows: true label, columns: prediction):

True \ predicted	neg	neu	pos	Total
neg	264	71	12	347
neu	213	1 184	169	1 566
pos	58	133	284	475
Total	535	1 388	465	2 388

A1: Attenuation from Misclassification (FinBERT) (task 2/2) [Solved]

Task

What to do

- ▶ (a) Write the probability limit of the OLS slope of r on $\hat{s}$ and show that it equals $\beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$ under the assumption above
- ▶ (b) Derive $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$ and $\text{Var}(\hat{s}) = p(1 - p)$, hence the attenuation factor $\lambda = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1) / [p(1 - p)]$
- ▶ (c) Compute π , p , α_0 and α_1 from the first row and the first column of the matrix, then λ to three decimal places
- ▶ (d) Compute $\text{Cov}(u, s^*)$ for the measurement error $u = \hat{s} - s^*$ and decide whether the error is classical (uncorrelated with s^*)

What to report

- ▶ λ and the factor $1/\lambda$ by which the true β exceeds the observed slope [Aigner \(1973\)](#)
- ▶ Interpretation: what does λ imply for a slope on the FinBERT negative-news dummy, and why does it not carry over to the continuous-score slopes of Lecture 15?

A1: Solution (1/2) [Solved]

Solution

- (a) For one regressor, $\text{plim } \hat{b} = \text{Cov}(r, \hat{s}) / \text{Var}(\hat{s})$; substituting $r = a + \beta s^* + \varepsilon$ gives $\text{Cov}(r, \hat{s}) = \beta \text{Cov}(s^*, \hat{s}) + \text{Cov}(\varepsilon, \hat{s})$
- The error is non-differential, so $\text{Cov}(\varepsilon, \hat{s}) = 0$ and $\text{plim } \hat{b} = \beta \text{Cov}(s^*, \hat{s}) / \text{Var}(\hat{s})$
- (b) Both variables are binary: $\text{Cov}(s^*, \hat{s}) = P(s^* = 1, \hat{s} = 1) - \pi p = \pi(1 - \alpha_1) - \pi p$
- By total probability $p = \pi(1 - \alpha_1) + (1 - \pi)\alpha_0$; substituting,
 $\text{Cov}(s^*, \hat{s}) = \pi(1 - \pi)(1 - \alpha_1) - \pi(1 - \pi)\alpha_0 = \pi(1 - \pi)(1 - \alpha_0 - \alpha_1)$
- $\hat{s}$ is a Bernoulli variable, so $\text{Var}(\hat{s}) = p(1 - p)$, and the ratio of the two gives λ

A1: Solution (2/2) [Solved]

Solution

- (c) $\pi = 347/2\,388 = 0.1453$ and $p = 535/2\,388 = 0.2240$
- $\alpha_1 = 1 - 264/347 = 0.2392$ (missed negatives); $\alpha_0 = 271/2\,041 = 0.1328$ (neutral or positive headlines called negative)
- $\lambda = \frac{0.1242 \times 0.6280}{0.2240 \times (1 - 0.2240)} = \frac{0.0780}{0.1738} = 0.449$, so $\beta = 2.23 \hat{b}$
- (d) $\text{Cov}(u, s^*) = \text{Cov}(\hat{s}, s^*) - \text{Var}(s^*) = 0.0780 - 0.1242 = -0.0462 < 0$: the error is negatively correlated with the truth, so it is not classical [Aigner \(1973\)](#)
- Interpretation: a slope on the FinBERT dummy keeps only about 45% of the true effect; the Lecture 15 slopes use a standardised continuous score, attenuated by $\rho(s^*, \hat{s})$ instead (lecture), so λ does not transfer

A2: Attenuation for the LM Dictionary (task 1/2) [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Same regression and the same non-differential assumption as in A1; now $\hat{s} = 1$ when the Loughran–McDonald (LM) tone of the headline is negative
- ▶ LM confusion matrix on the same headlines (rows: true label, columns: prediction):

True \ predicted	neg	neu	pos	Total
neg	172	166	9	347
neu	247	1 192	127	1 566
pos	45	349	81	475
Total	464	1 707	217	2 388

A2: Attenuation for the LM Dictionary (task 2/2) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) Compute π , p , α_0 and α_1 from the first row and the first column of the matrix
- ▶ (b) Compute the attenuation factor λ and the correction factor $1/\lambda$ with the formula of A1
- ▶ (c) Compute $\text{Cov}(u, s^*)$ and say whether the LM error is classical
- ▶ (d) Compare λ with the FinBERT value of A1: which classifier attenuates more, and by what ratio?

What to report

- ▶ The four rates, λ , $1/\lambda$ and the ratio of the two factors
- ▶ Interpretation: is the weaker same-day LM slope of Lecture 15 (15.5 bp against 39.2 bp for FinBERT) evidence of a weaker market reaction to LM-negative news? Model: A1

A3: Paired Accuracy Difference: Variance, Exact Test, Power (task 1/2) [Solved]

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = LM dictionary, classifier B = FinBERT; discordant counts $n_{01} = 539$ (A wrong, B right) and $n_{10} = 252$ (A right, B wrong); headlines are independent
- ▶ For headline i , $D_i = 1$ if A is wrong and B right, $D_i = -1$ if A is right and B wrong, $D_i = 0$ otherwise; the accuracy difference is $d = ED_i = \text{acc}_B - \text{acc}_A$

A3: Paired Accuracy Difference: Variance, Exact Test, Power (task 2/2) [Solved]

Task

□ What to do

- ▶ (a) Show that $\hat{d} = (n_{01} - n_{10})/n$ and derive $\text{Var}(\hat{d}) = (q - d^2)/n$, where $q = P(D_i \neq 0)$ is estimated by $(n_{01} + n_{10})/n$
- ▶ (b) Compute $\hat{d}$, its standard error, the 95% CI $\hat{d} \pm 1.96 \text{ SE}$ and $z = \hat{d}/\text{SE}$
- ▶ (c) Run the exact McNemar test [McNemar \(1947\)](#): under $H_0 : d = 0$, $n_{01} \sim \text{Bin}(n_{01} + n_{10}, 0.5)$; give the two-sided p -value and the decision at $\alpha = 5\%$
- ▶ (d) Approximate the power at $n = 2388$ if the true d equals $\hat{d}$, $\text{power} \approx \Phi(|d|\sqrt{n}/\sqrt{q - d^2} - 1.96)$, and the n needed for 80% power, $n_{80} = (1.96 + 0.84)^2(q - d^2)/d^2$

□ What to report

- ▶ $\hat{d}$, SE, CI, z , exact p , the decision, the power and n_{80} ; compare the CI with the paired bootstrap CI of B1
- ▶ Interpretation: which headlines determine the variance of $\hat{d}$?

A3: Solution (1/2) [Solved]

Solution

- (a) $\hat{d} = \frac{1}{n} \sum_i D_i = (n_{01} - n_{10})/n$, because 539 terms equal +1, 252 equal -1 and the rest are 0
- $D_i^2 = 1$ exactly when one classifier is right and the other wrong (two wrong but different labels give $D_i = 0$), so $ED_i^2 = q$ and $\text{Var}(D_i) = q - d^2$; with independent headlines, $\text{Var}(\hat{d}) = (q - d^2)/n$
- (b) $\hat{d} = (539 - 252)/2\,388 = 0.1202$, $\hat{q} = 791/2\,388 = 0.3312$
- $SE = \sqrt{(0.3312 - 0.01444)/2\,388} = 1.15 \text{ pp}$; CI 95% [9.76; 14.28] pp, close to the bootstrap CI [+9.8; +14.2] of B1; $z = 10.43$

A3: Solution (2/2) [Solved]

Solution

- ▣ (c) Under H_0 we expect 395.5 of the 791 discordant headlines on each side; observing 539 gives an exact two-sided $p < 0.001$
- ▣ Decision: reject H_0 at 5%; FinBERT is more accurate than LM on these headlines
- ▣ (d) $|d|\sqrt{n}/\sqrt{q-d^2} - 1.96 = +8.47$, so the power is $\Phi(+8.47) \approx 100\%$; $n_{80} = 7.84 \times (0.3312 - 0.01444)/0.01444 \approx 170$ headlines
- ▣ Interpretation: only the discordant headlines (q) enter the variance; a 12-point gap is detectable with a few hundred headlines

A4: Is “No Difference” a Power Problem? [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = FinBERT, classifier B = Qwen2.5-7B zero-shot (prompt P1); $n_{01} = 395$ (A wrong, B right), $n_{10} = 355$ (A right, B wrong)
- ▶ For headline i , $D_i = 1$ if A is wrong and B right, $D_i = -1$ if A is right and B wrong, $D_i = 0$ otherwise; the accuracy difference is $d = ED_i = \text{acc}_B - \text{acc}_A$

▣ What to do

- ▶ (a) Compute $\hat{d}$, $\hat{q}$, SE, the 95% CI and z with the formulas of A3
- ▶ (b) Run the exact McNemar test at $\alpha = 5\%$ and state the decision
- ▶ (c) Compute the power at $n = 2\,388$ if the true d equals $\hat{d}$, and n_{80}
- ▶ (d) Use the CI to state which differences in accuracy the data rule out

▣ What to report

- ▶ All the numbers of (a)–(c); interpretation: may you conclude that the two models are equally accurate? Model: A3

A5 (1/2): From Label Probabilities to a Score [Solved]

Task

□ Context and assumptions

- ▶ Qwen2.5-7B, zero-shot with prompt P1, one Twitter validation headline that the annotators labelled **neutral**
- ▶ Answer-word probabilities (rounded): $p_{\text{neg}} = 0.608$, $p_{\text{neu}} = 0.392$, $p_{\text{pos}} = 0.000$ (exactly 1.0×10^{-12}); they come from logits by $p_k = e^{z_k/T} / \sum_j e^{z_j/T}$ with $T = 1$

□ What to do

- ▶ (a) Give the predicted label $\arg \max_k p_k$ and the net score $s = p_{\text{pos}} - p_{\text{neg}}$, and say whether the label is right
- ▶ (b) Compute the logit gap $z_{\text{neu}} - z_{\text{neg}} = \ln(p_{\text{neu}}/p_{\text{neg}})$
- ▶ (c) Compute the probabilities, the label and the score at $T = 2$, using $p_k(T) \propto p_k^{1/T}$

□ What to report

- ▶ The label, s , the logit gap and $p(T)$, $s(T)$; interpretation: what happens to a trading rule that uses a threshold on s when T changes?

A5 (1/2): Solution [Solved]

Solution

- ▣ (a) The largest probability is $p_{\text{neg}} = 0.608$: the label is negative, which is wrong (the truth is neutral)
- ▣ $s = p_{\text{pos}} - p_{\text{neg}} = 0.000 - 0.608 = -0.608$: a strongly negative score for a neutral headline
- ▣ (b) $z_{\text{neu}} - z_{\text{neg}} = \ln(0.392/0.608) = -0.437$: neutral is behind by less than half a logit
- ▣ (c) With $T = 2$, $p_k(T) \propto \sqrt{p_k}$: $\sqrt{p} = (0.780, 0.626, 1.0 \times 10^{-6})$ with sum 1.406, so $p(T) = (0.554, 0.446, 0.000)$
- ▣ The label stays negative and $s(T) = -0.554$; interpretation: temperature keeps the ranking but pulls the score towards 0, so a rule such as $|s| > 0.5$ selects different headlines

A5 (2/2): Fitting the Temperature [Solved]

Task

□ Context and assumptions

- ▶ Qwen2.5-7B probabilities on a random half of the Twitter validation set ($n = 1\,194$ headlines, seed 15); logits $z_{ik} = \ln p_{ik}$, true label y_i
- ▶ Rescaled probabilities $p_{ik}(T) = e^{z_{ik}/T} / \sum_l e^{z_{il}/T}$, $T > 0$; loss $\text{NLL}(T) = -\sum_i \ln p_{iy_i}(T)$ Guo et al. (2017)

□ What to do

- ▶ (a) Explain why the predicted labels and the accuracy do not depend on T , while the Brier score does
- ▶ (b) Derive $\text{NLL}'(T)$ and write the first-order condition for the optimal T
- ▶ (c) Evaluate the average bracket of the condition at $T = 1$, then solve for T^* with a root finder (Brent's method on $[0.5, 50]$)
- ▶ (d) On the other half, compute the ECE (10 equal-width bins of the top probability on $[1/3, 1]$) and the Brier score $\frac{1}{n} \sum_i \sum_k (p_{ik}(T) - 1\{y_i = k\})^2$ (summed over the 3 classes) before and after scaling

□ What to report

- ▶ T^* , the sign at $T = 1$, ECE and Brier before and after; interpretation: is the LLM over- or under-confident?

A5 (2/2): Solution [Solved]

Solution

- (a) $z \mapsto z/T$ is increasing for $T > 0$, so the order of z_{i1}, z_{i2}, z_{i3} and hence $\arg \max$ and accuracy do not change; the probabilities change, and with them Brier and log loss
- (b) $\ln p_{iy}(T) = z_{iy}/T - \ln \sum_k e^{z_{ik}/T}$, so $\frac{d}{dT} \ln p_{iy}(T) = -\frac{z_{iy}}{T^2} + \frac{1}{T^2} \sum_k p_{ik}(T) z_{ik}$
- $\text{NLL}'(T) = \frac{1}{T^2} \sum_i (z_{iy_i} - \sum_k p_{ik}(T) z_{ik}) = 0$: on average the true-class logit must equal its expected value under the model
- (c) At $T = 1$ the average bracket is $-2.47 < 0$, so NLL decreases as T grows; the root is $T^* = 8.43$
- (d) ECE falls from 22.6% to 4.3% and the Brier score from 0.468 to 0.366; interpretation: the raw LLM probabilities are heavily overconfident

A6: Fleiss' κ and One Dawid–Skene Step (task 1/2) [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Illustrative table (numbers made up for the exercise): 6 sentences, $m = 5$ raters each, votes for negative, neutral, positive
- ▶ Fleiss' κ [Fleiss \(1971\)](#): $P_i = \sum_j n_{ij}(n_{ij} - 1)/[m(m - 1)]$, $\bar{P} = \frac{1}{N} \sum_i P_i$, $p_j =$ share of all votes for j , $\bar{P}_e = \sum_j p_j^2$,
 $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$

Sentence	neg	neu	pos
1	5	0	0
2	0	5	0
3	0	1	4
4	1	3	1
5	2	3	0
6	0	2	3

A6: Fleiss' κ and One Dawid–Skene Step (task 2/2) [Proposed]

Solution discussed in class

Task

□ What to do

- ▶ (a) Compute P_i for every sentence and $\bar{P}$
- ▶ (b) Compute p_j , $\bar{P}_e$ and κ
- ▶ (c) E-step of [Dawid & Skene \(1979\)](#): assume every rater gives the true class with probability 0.8 and each other class with probability 0.1, and a uniform prior; compute $P(\text{true} = j \mid \text{votes}) \propto \prod_l \pi_{jl}^{n_{jl}}$ for sentences 4 and 5
- ▶ (d) Compare the posterior with the vote shares and describe the M-step

□ What to report

- ▶ κ and the two posteriors; interpretation: how would you treat the disputed sentences in a test set, and why? Model: A1 (misclassification probabilities)

A7: Break-Even Cost with a Standard Error [Solved]

Task

▣ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ FinBERT signal: mean gross return over all days $\hat{\mu} = 3.27$ bp, Newey–West $t = 1.27$ Newey & West (1987) with $L = 8$ lags ($L = \lfloor 4(n/100)^{2/9} \rfloor$), share of days with positions $a = 0.962$, gross Sharpe $\sqrt{252} \hat{\mu} / \sigma = 0.36$

▣ What to do

- ▶ (a) Write the mean net daily return and the annualised net Sharpe ratio as functions of the cost c per trade, assuming the costs barely change the volatility σ
- ▶ (b) Derive the break-even cost c^* at which the mean net return is zero
- ▶ (c) Compute $\text{SE}(\hat{\mu}) = \hat{\mu} / t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI $c^* \pm 1.96 \text{SE}$
- ▶ (d) Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp

▣ What to report

- ▶ c^* , SE, CI and the net Sharpe ratio; interpretation: is a realistic cost of 5 bp per trade inside the CI of c^* ?

A7: Solution [Solved]

Solution

- (a) On a fraction a of the days we pay $4c$, so $\mu_{\text{net}} = \mu - 4ac$ and $SR_{\text{net}} = \sqrt{252}(\mu - 4ac)/\sigma$
- (b) $\mu - 4ac^* = 0$ gives $c^* = \mu/(4a) = 3.27/3.85 = 0.85$ bp per trade
- (c) $SE(\hat{\mu}) = 3.27/1.27 = 2.57$ bp; treating a as fixed (the activity schedule is set by the news calendar), c^* is linear in $\hat{\mu}$, so $SE(c^*) = 2.57/3.85 = 0.67$ bp and the CI is $[-0.46; 2.16]$ bp
- (d) $\sigma = \sqrt{252} \hat{\mu} / SR = \sqrt{252} \times 3.27 / 0.36 = 143$ bp; $SR_{\text{net}} = \sqrt{252}(3.27 - 3.85 \times 5) / 143 = -1.77$
- Interpretation: 5 bp lies far above the upper bound of the CI; with daily turnover the strategy cannot survive realistic costs

A8: The Qwen Strategy after Costs [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ Qwen2.5-7B signal: $\hat{\mu} = 5.64$ bp, Newey–West $t = 2.26$ ($L = 8$), $a = 0.950$, gross Sharpe 0.63

▣ What to do

- ▶ (b) Derive the break-even cost c^* at which the mean net return is zero
- ▶ (c) Compute $\text{SE}(\hat{\mu}) = \hat{\mu}/t$, then $\text{SE}(c^*)$ by the delta method and the 95% CI $c^* \pm 1.96 \text{ SE}$
- ▶ (d) Recover σ from the gross Sharpe ratio and compute the net Sharpe ratio at $c = 5$ bp
- ▶ (e) Lecture 15 reports a break-even cost of 1.5 bp and a net Sharpe ratio of -1.48 : decide whether this is a copy error

▣ What to report

- ▶ c^* , SE, CI, net Sharpe; interpretation: does the significant gross mean make the strategy tradable? Model: A7

B1: Is FinBERT Better than LM? [Solved]

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = LM dictionary (label = sign of the tone), classifier B = FinBERT (label = largest probability); code: `b_pairs` in the notebook

▣ What to do

- ▶ (a) Compute the accuracy of each classifier with a percentile bootstrap 95% CI: resample the headlines with replacement 2 000 times (seed 0) and take the 2.5% and 97.5% quantiles
- ▶ (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers in each of the 2 000 draws
- ▶ (c) Count n_{01} and n_{10} , run the exact McNemar test and the asymptotic version $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ_1^2 (critical value 3.84) at $\alpha = 5\%$

▣ What to report

- ▶ A table with the accuracies, CIs and the difference; n_{01} , n_{10} , p -values and the decision
- ▶ Interpretation: why is the paired test more powerful than comparing the two intervals, which headlines carry no information, and is the gap useful for trading?

B1: Solution (1/2) [Solved]

	Accuracy (%)	CI 95%
LM	60.5	[58.5; 62.4]
FinBERT	72.5	[70.7; 74.4]
Difference (pp)	+12.0	[+9.8; +14.2]

Solution

- Step 1: labels of both classifiers on the 2388 headlines; step 2: 2000 bootstrap resamples of the headline indices; step 3: accuracies and their difference in each resample
- (a)–(b) The intervals of the two classifiers do not overlap and the paired CI of the difference excludes 0

B1: Solution (2/2) [Solved]

Solution

- ▣ (c) $n_{01} = 539$ (LM wrong, FinBERT right), $n_{10} = 252$; under H_0 we expect 395.5 each; exact binomial $p < 0.001$
- ▣ Asymptotic: $(539 - 252)^2 / 791 = 104.1 > 3.84$; decision: reject equal accuracy at 5%
- ▣ Pairing removes the variation of headline difficulty: headlines that both get right or both get wrong cancel out and carry no information
- ▣ Interpretation: FinBERT is clearly better than the dictionary outside its own training data; a 12-point gain in labelling accuracy does not yet say anything about returns (B5)

B2: Is a 7-Billion-Parameter LLM Better than FinBERT? [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Classifier A = FinBERT, classifier B = Qwen2.5-7B zero-shot with prompt P1; neither used any Twitter label

▣ What to do

- ▶ (a) Compute the accuracy of each classifier with a percentile bootstrap 95% CI: resample the headlines with replacement 2 000 times (seed 0) and take the 2.5% and 97.5% quantiles
- ▶ (b) Compute the difference $B - A$ with a paired bootstrap CI: the same resampled headlines for both classifiers in each of the 2 000 draws
- ▶ (c) Count n_{01} and n_{10} , run the exact McNemar test and the asymptotic version $(n_{01} - n_{10})^2 / (n_{01} + n_{10})$ against χ^2_1 (critical value 3.84) at $\alpha = 5\%$

▣ What to report

- ▶ The same table and tests as in B1; interpretation: do the two intervals overlap, does that settle the question, and what would you still gain from a much larger and slower model? Model: B1

B3: Training-Data Leakage [Solved]

Task

▣ Context and assumptions

- ▶ Financial PhraseBank [Malo et al. \(2014\)](#): 4 846 news sentences (2 264 with full annotator agreement); FinBERT was fine-tuned on a training split of these sentences (3 101 of 4 845 in [Araci \(2019\)](#))
- ▶ Twitter validation set, 2 388 headlines, unseen by FinBERT; four classifiers: FinBERT, Qwen2.5-7B zero-shot, LM, MiniLM embeddings + logistic regression trained on the Twitter training set [Wang et al. \(2020\)](#)

▣ What to do

- ▶ (a) Compute accuracy (with a 2 000-draw bootstrap CI) and macro-F1 of each classifier on PhraseBank with full agreement, on all of PhraseBank and on Twitter
- ▶ (b) Compute the drop from all of PhraseBank to Twitter for each classifier
- ▶ (c) Compute the difference in differences FinBERT minus Qwen2.5-7B and its bootstrap CI, resampling each data set independently 2 000 times

▣ What to report

- ▶ A table of accuracies, drops and macro-F1; the difference in differences with its CI
- ▶ Interpretation: how do you separate “harder data” from “memory of the training set”, and why does MiniLM move the other way?

B3: Solution [Solved]

Accuracy (%)	PhraseBank, 100% agr.	PhraseBank, all	Twitter	Drop (pp)	Macro-F1, Twitter
FinBERT	97.2	89.0	72.5	16.4	66.8
Qwen2.5-7B	93.6	81.4	74.2	7.2	72.6
LM	65.0	62.5	60.5	2.0	46.2
MiniLM + log. reg.	81.4	70.6	78.6	-8.0	69.9

Solution

- (b) Drop = all of PhraseBank minus Twitter; FinBERT CI on PhraseBank [88.1; 89.8], on Twitter [70.7; 74.4]
- (c) $(89.0 - 72.5) - (81.4 - 74.2)$, computed from unrounded accuracies: $16.4 - 7.2 = 9.2$ pp, bootstrap CI [6.7; 11.9]: it excludes 0
- Interpretation: Qwen2.5-7B was not fine-tuned on PhraseBank labels, so its drop is the benchmark for the harder data; if both would lose the same without training overlap (an untested assumption), FinBERT's extra 9.2 points come from overlap; the CI covers sampling noise only; MiniLM is trained on Twitter, so it is better at home

B4: How Many Labels Do You Need? [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Twitter Financial News: training set of 9 543 labelled headlines, validation set of 2 388; benchmark FinBERT at 72.5% accuracy, which uses no Twitter label
- ▶ Two supervised models: logistic regression ($C = 4$) on 384-dimensional MiniLM embeddings, and logistic regression ($C = 8$) on TF-IDF features of words and word pairs

▣ What to do

- ▶ (a) For $n \in \{100, 200, 400, \dots, 6\,400, 9\,543\}$ draw 10 random training subsets of size n (one for the full set), fit both models and compute the mean validation accuracy
- ▶ (b) Find the smallest n at which each model beats FinBERT on average
- ▶ (c) Report the standard deviation of the embedding accuracy across the 10 repetitions at $n = 100$

▣ What to report

- ▶ The learning-curve table; interpretation: why do embeddings start higher but grow more slowly, and how many texts would you label for Romanian stock news? Model: B3

B5: Reaction or Prediction, and Which Standard Errors? [Solved]

Task

▣ Context and assumptions

- ▶ Panel of 18 282 stock-days with news on 16 FNSPID stocks, 3 434 trading days, 27.10.2009–18.12.2023; daily score = mean of the headline scores, standardised over the panel (z)
- ▶ Returns: daily adjusted closes of the stocks and of SPY from data/market; excess return r^e = stock minus SPY, in %; day d = first trading day on or after the headline date, so $d + 2$ is the first return surely after publication

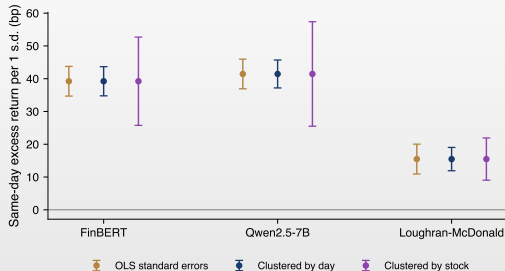
▣ What to do

- ▶ (a) Estimate by OLS $r_{i,d+h}^e = a + b z_{i,d} + \varepsilon$ for FinBERT, Qwen2.5-7B and LM and $h = 0, 1, 2$ (nine regressions); report b in bp per one standard deviation of the score
- ▶ (b) Compute three standard errors: OLS, clustered by day ($G = 3\,434$) and clustered by stock ($G = 16$), with the sandwich estimator and the small-sample factor $\frac{G}{G-1} \frac{n-1}{n-2}$ [Petersen \(2009\)](#)
- ▶ (c) Compute the t -statistics and compare them with 1.96 and with the t_{15} critical value 2.13 for the stock clustering

▣ What to report

- ▶ A chart of the $h = 0$ slopes with the three intervals and a table of all nine regressions
- ▶ Interpretation: why do $h = 0$ and $h = 2$ differ, which clustering matters, and can you trust a clustered SE with 16 clusters?

B5: Solution (1/3) [Solved]



Solution

- Same-day slope b (bp per one standard deviation of score) with 95% intervals from the three standard errors: the stock-clustered interval is much wider

B5: Solution (2/3) [Solved]

Score	h	b (bp)	SE OLS	SE day	SE stock	t day	t stock
FinBERT	0	+39.2	2.3	2.3	6.9	+17.27	+5.71
FinBERT	2	-0.5	2.2	2.2	1.5	-0.25	-0.35
Qwen2.5-7B	0	+41.4	2.3	2.2	8.1	+19.07	+5.10
Qwen2.5-7B	2	+2.3	2.2	1.9	1.0	+1.19	+2.41
LM	0	+15.5	2.3	1.8	3.3	+8.53	+4.70
LM	2	-0.0	2.2	1.7	0.9	-0.01	-0.03

Solution

- $h = 0$: large under every standard error (t by stock above 4.7): headlines describe the move of the day, often the move itself ("shares jump")
- $h = 2$: FinBERT -0.5 bp ($t = -0.25$ by day): the tradable forecast is not there

B5: Solution (3/3) [Solved]

Solution

- Clustering by day is close to OLS: with only 5.3 stocks per day there is little common noise to absorb
- Clustering by stock multiplies the FinBERT SE at $h = 0$ by 3.0: the slope differs across stocks, and each stock's errors are correlated over time
- With 16 clusters the clustered SE is itself noisy: Qwen at $h = 2$ has $t = +2.41$ by stock, $p = 0.029$ with t_{15} , but 0.061 by wild cluster bootstrap [Cameron, Gelbach & Miller \(2008\)](#) and 0.37 after Holm over the nine regressions (B10)
- Interpretation: sentiment explains the news day, not the tradable days; with 16 stocks, inference must use few-cluster methods

B6: Event Study with the Loughran–McDonald Tone (task 1/2) [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Panel of 18 282 stock-days with news on 16 FNSPID stocks, 3 434 trading days, 27.10.2009–18.12.2023; daily score = mean of the headline scores, standardised over the panel (z)
- ▶ Returns: daily adjusted closes of the stocks and of SPY from data/market; excess return r^e = stock minus SPY, in %; day d = first trading day on or after the headline date, so $d + 2$ is the first return surely after publication

B6: Event Study with the Loughran–McDonald Tone (task 2/2) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) The daily LM tone is exactly 0 on 48% of stock-days, so both tercile cut-offs are 0: form the disjoint groups $\text{tone} < 0$ (5 196 stock-days) and $\text{tone} > 0$ (4 288), dropping the zeros
- ▶ (b) For each date, average across that date's stock-days the excess returns on days $-5, \dots, +5$ around the news day
- ▶ (c) Compute the mean over dates of the CAR (cumulative abnormal return) for $[-5, -1]$, day 0, day 1 and $[+2, +5]$, with Newey–West t (lags $\lfloor 4(n/100)^{2/9} \rfloor$)
- ▶ (d) Repeat (b)–(c) for the bottom and top terciles of the FinBERT score (no ties there)

What to report

- ▶ A table of the four windows for both scores and both groups; interpretation: does LM separate winners from losers on day 0 as well as FinBERT, and why is the post-event return positive for both groups [Tetlock \(2007\)](#)? Model: B5

B7: Does the Model Remember the Market? [Solved]

Task

▣ Context and assumptions

- ▶ S&P 500 monthly closes from data/market (GSPC.INDX), January 2000 to August 2026; a month is “up” if its return is positive
- ▶ Qwen2.5 1.5B, 7B and 14B, weights published on 19 September 2024 [Qwen Team \(2024\)](#), asked without any context “Did the S&P 500 index rise or fall during ⟨month year⟩?”; score $P(\text{rise})$
- ▶ Before: 296 months to August 2024 (185 up); after: 23 months from October 2024 (13 up)

▣ What to do

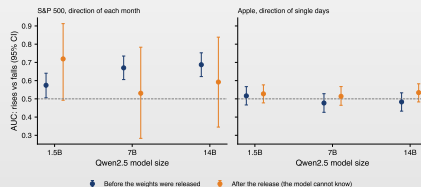
- ▶ (a) Compute the $\text{AUC} = U/(n_{\text{up}}n_{\text{down}})$ (Mann–Whitney U) for each model and period, with a 2000-draw bootstrap CI over months
- ▶ (b) Run a permutation test for 14B before the release: shuffle the up/down labels 2000 times; p = share of shuffled AUCs at least as large as the observed one
- ▶ (c) Test $\text{AUC}_{\text{before}} = \text{AUC}_{\text{after}}$ with DeLong variances [DeLong et al. \(1988\)](#) (independent samples) at 5%, and compute the smallest drop detectable with 80% power

▣ What to report

- ▶ A table of AUCs with CIs and a chart; the permutation p , the DeLong z and p , the detectable drop
- ▶ Interpretation: what does an AUC above 0.5 mean when the model sees no data, and what does it imply for backtests on old news [Glasserman & Lin \(2023\)](#)?

B7: Solution (1/2) [Solved]

Qwen2.5	AUC before	CI 95%	AUC after	CI 95%	$\bar{P}$ (rise) up / down
1.5B	0.57	[0.51; 0.64]	0.72	[0.49; 0.91]	0.36 / 0.30
7B	0.67	[0.61; 0.73]	0.53	[0.28; 0.78]	0.73 / 0.43
14B	0.69	[0.62; 0.75]	0.59	[0.35; 0.84]	0.81 / 0.54



Solution

- (a) 7B and 14B rank up months above down months well above chance before the release; the 1.5B model barely

B7: Solution (2/2) [Solved]

Solution

- (b) Permutation p for 14B before the release < 0.001 (2000 shuffles): reject “no knowledge” at 5%
- (c) DeLong: SE before 0.032, after 0.126; $z = 0.73$, $p = 0.47$: do not reject equality
- Under H_0 the post-release SE is 0.124, so the smallest drop detectable with 80% power is $2.80 \times \sqrt{SE_{\text{before}}^2 + SE_0^2} = 0.36$; about 101 post-release months would be needed
- Interpretation: no data were given, so the discrimination most plausibly comes from the training text (outcomes or the regimes around them; months treated as exchangeable in the bootstrap and permutation); the post-release sample is far too short to show that the memory is gone, and a backtest on pre-release news mixes forecasting with remembering

B8: Prompt Sensitivity [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Qwen2.5-7B and 14B with three prompts that ask the same question: P1 “classify the sentiment . . . for investors”, P2 “good, bad or neutral for the stock price”, P3 “what is the sentiment of this text”

▣ What to do

- ▶ (a) Compute the accuracy with a 2 000-draw bootstrap CI and the share of “neutral” answers for each prompt and model
- ▶ (b) Run the exact McNemar test of P1 against P3 for each model at 5%
- ▶ (c) Compute the share of headlines that get the same label under all three prompts

▣ What to report

- ▶ A table with the three prompts and the test; interpretation: if you picked the best prompt on this set, what would you be doing, and why does the best prompt also answer “neutral” most often? Model: B1

B9: Prediction-Powered Inference with a Few Human Labels [Solved]

Task

▣ Context and assumptions

- ▶ Data: Twitter Financial News, validation set of $N = 2\,388$ headlines published in 2022, each labelled negative, neutral or positive by annotators; the human label is treated as the truth
- ▶ Target $\theta =$ mean human net tone $(-1, 0, 1)$ of the headline population, represented by the 2 388 headlines; model score $f =$ net score $P(\text{pos}) - P(\text{neg})$ of FinBERT or of Qwen2.5-7B
- ▶ Design: n labelled headlines (with Y) and $N - n$ unlabelled headlines (only f) are drawn independently, with replacement, from the 2 388; f is fixed (not trained on these headlines)

▣ What to do

- ▶ (a) For one draw with $n = 200$, compute the labels-only estimate $\bar{Y}_n \pm 1.96 s_Y / \sqrt{n}$
- ▶ (b) Compute PPI [Angelopoulos et al. \(2023\)](#): $\hat{\theta} = \frac{1}{N-n} \sum_{\text{unlab.}} f - \frac{1}{n} \sum_{\text{lab.}} (f - Y)$ with SE $= \sqrt{\hat{V}(f)/(N-n) + \hat{V}(f - Y)/n}$, for both models
- ▶ (c) For $n = 100, 200, 400, 800$, draw 2 000 times n labelled and $N - n$ unlabelled headlines independently, with replacement (the 2 388 act as the population), and record the coverage of θ and the mean width of each 95% CI

▣ What to report

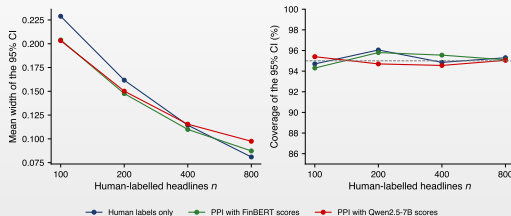
- ▶ The one-draw estimates, a coverage and width table or chart; interpretation: why is the plain model mean $\frac{1}{N} \sum f$ not an estimate of θ , when is PPI narrower, and how would you use it for the FNSPID regressions?

B9: Solution (1/2) [Solved]

Solution

- ▣ True $\theta = 0.054$; plain model means: FinBERT -0.014 (wrong sign), Qwen $+0.101$: the model bias does not shrink as N grows
- ▣ (a) One draw, $n = 200$: labels only $0.010 \pm 1.96 \times 0.042$
- ▣ (b) PPI-FinBERT: rectifier $\frac{1}{n} \sum (f - Y) = -0.064$, so $\hat{\theta} = 0.054 \pm 1.96 \times 0.036$; PPI-Qwen: rectifier -0.012 , $\hat{\theta} = 0.123 \pm 1.96 \times 0.039$
- ▣ The rectifier removes the bias in expectation, but in one draw it can move an estimate away from θ (Qwen here); all three intervals contain the true value in this draw

B9: Solution (2/2) [Solved]



Solution

- (c) $n = 100$: coverage 94.7%, 94.3%, 95.4%, width 0.229, 0.204, 0.203 (labels, PPI-FinBERT, PPI-Qwen)
- PPI is narrower when $\text{Var}(f)/(N - n) + \text{Var}(f - Y)/n < \text{Var}(Y)/n$; for large N the gain is $1 - \text{Var}(f - Y)/\text{Var}(Y)$: 24% (FinBERT), 27% (Qwen); at $n = 800$ only $N - n = 1\,588$ texts are unlabelled and PPI is wider (0.087 vs 0.081); coverage stays near 95%
- Interpretation: for FNSPID, label all headlines of a random sample of stock-days and rectify every OLS moment condition that contains the score (C1, part (e))

B10: Wild Cluster Bootstrap with 16 Stocks [Proposed]

Solution discussed in class

Task

Context and assumptions

- ▶ Panel of 18 282 stock-days with news on 16 FNSPID stocks, 3 434 trading days, 27.10.2009–18.12.2023; daily score = mean of the headline scores, standardised over the panel (z)
- ▶ The nine regressions of B5 (three scores $\times h = 0, 1, 2$), with standard errors clustered by stock ($G = 16$); one-way clustering assumes independent stocks

What to do

- ▶ (a) Compute the p -value of each slope with t_{15} critical values
- ▶ (b) Run the wild cluster bootstrap with the null imposed [Cameron, Gelbach & Miller \(2008\)](#): $y_i^* = \bar{y} + w_{g(i)}\tilde{\varepsilon}_i$ with restricted residuals $\tilde{\varepsilon}_i = y_i - \bar{y}$, Rademacher weights $w_g = \pm 1$, 9 999 draws; $p = P^*(|t^*| \geq |t|)$
- ▶ (c) Compute t with two-way clustering by day and stock, $\hat{V} = \hat{V}_{\text{day}} + \hat{V}_{\text{stock}} - \hat{V}_{\text{day} \cap \text{stock}}$
- ▶ (d) Adjust the nine bootstrap p -values with Holm's step-down procedure at 5%

What to report

- ▶ A table with t , the three p -values and the two-way t ; interpretation: why is $N(0, 1)$ a poor reference for a t built from 16 clusters, and does Qwen at $h = 2$ survive? Model: B5

B11: Specification Curve for the Sentiment Strategy [Proposed]

Solution discussed in class

Task

□ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ Grid of Lecture 15: three scores (LM, FinBERT, Qwen) $\times$ thresholds $|s| > b$, $b \in \{0, 0.25, 0.5\}$, $\times$ return day $d + 2, \dots, d + 5$ (each a one-day position opened at the previous close): 36 strategies

□ What to do

- ▶ (a) Compute the mean daily return and its Newey–West t for each of the 36 strategies, and plot the t -statistics sorted [Simonsohn, Simmons & Nelson \(2020\)](#)
- ▶ (b) Simulate the joint null of no predictability: flip the signs of the positions at random in blocks of 10 consecutive signal days (dependent wild bootstrap [Shao \(2010\)](#)), the same flips for all 36 strategies, 2000 times
- ▶ (c) Compute the joint p -values of the number of $t > 1.96$ and of the median t

□ What to report

- ▶ The sorted curve and the two joint p -values; interpretation: why must the flip be shared by all specifications, and what would a “best specification” report hide? Model: B7 (a randomisation test)

C1: Does Headline Sentiment Pay after Costs? (task 1/2) [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Strategy of Lecture 15 on the 16 FNSPID stocks, 3568 trading days: at the close of $d + 1$ take $\text{sign}(s_{i,d})$ in every stock with news on day d , hedged with SPY, close one day later; four trades on each day with positions
- ▶ Positions are hedged with SPY and equally weighted; days without news: no position and return 0; costs of 5 bp per trade, four trades on each day with positions
- ▶ Scores: FinBERT, Qwen2.5-7B and LM, from the saved tables of Lecture 15; returns from data/market

C1: Does Headline Sentiment Pay after Costs? (task 2/2) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) Compute the mean daily gross return, its Newey–West t and a moving-block bootstrap 95% CI (blocks of 10 days, 2 000 draws) for the FinBERT signal
- ▶ (b) Compute the annualised Sharpe ratio before and after costs, the break-even cost and the Sharpe ratio for costs of 0, 1, 2, 5 and 10 bp
- ▶ (c) Restrict to strong news, $|s_{i,d}| > 0.5$ (exploratory), and (d) repeat everything for Qwen2.5-7B and LM; the primary family, fixed in advance, is the three unfiltered strategies (Holm over them); if you select among all six variants, adjust over all six
- ▶ (e) Correct the same-day slope: label all headlines of a random sample of stock-days (true daily tone s^*); either divide b by $\rho(s^*, \hat{s})$ estimated there (the lecture correction for a standardised score; A1 applies only to a dummy), or rectify both OLS moments $E[(1, s^*)^\top (r - a - \beta s^*)] = 0$ with the score (B9, lecture) and multiply $\hat{\beta}$ by $\widehat{SD}(s^*)$, so both are effects of one SD of true tone; bootstrap over days, re-estimating ρ and $SD(s^*)$ in each draw
- ▶ (f) Report headlines after the Qwen2.5 release separately once available

What to report

- ▶ A two-page note with a results table and charts, and a Quantlet folder (code, Metainfo, charts); discuss timing, look-ahead of the LLM (B7), survivorship of the 16 stocks and crowding [Lopez-Lira & Tang \(2026\)](#)



C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

- **Student prompt:** "I have FNSPID headlines for 16 US stocks, 2009–2023, and daily excess returns vs SPY. Check that Qwen2.5-7B reads sentiment, then test whether its headline sentiment forecasts returns out of sample."
- **Answer of the AI assistant** (qwen_label, qwen_score: the zero-shot scorer of Lecture 15)

```
import pandas as pd, statsmodels.formula.api as smf
from sklearn.metrics import accuracy_score
tw['pred'] = [qwen_label(t) for t in tw['text']] # Twitter validation set
acc = accuracy_score(tw['label'], tw['pred'])
news['day'] = first_trading_day_on_or_after(news['date'])
news['s'] = [qwen_score(t) for t in news['title']] # P(pos) - P(neg)
d = news.groupby(['ticker', 'day'])['s'].mean().reset_index()
d['z'] = (d['s'] - d['s'].mean()) / d['s'].std()
d = d.merge(rets, on=['ticker', 'day']) # rets: ticker, day, xret (%)
fit = smf.ols('xret ~ z', data=d).fit(cov_type='cluster',
                                     cov_kws={'groups': pd.factorize(d['day'])[0]})
```

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

□ Conclusion of the AI assistant

- ▶ “Accuracy 74.2%, more than twice the 33% of a random guess among three classes: Qwen2.5-7B reads financial sentiment very well.”
 - ▶ “Slope +41.4 bp per standard deviation of the score, $t = +19.07$ (clustered by day): the day's headlines forecast the stock's excess return.”
 - ▶ “Qwen2.5 was never trained on these labels or return series, so the 2009–2023 backtest is fully out of sample.”
- The answer contains **three** errors typical for sentiment backtests; the code runs and the numbers are right
- ▶ (a) Find each error: name the line of code or the sentence, and explain why it is wrong
 - ▶ (b) Say how you would detect it without knowing the answer: a class count, a timestamp check, a release date, a memory test
 - ▶ (c) Write the corrected code or the corrected statement, with the numbers it produces
 - ▶ (d) Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

C3: Replicating Lopez-Lira and Tang on FNSPID (task 1/3) [Proposed]

Solution discussed in class

Task

▣ Context and assumptions

- ▶ Lecture 15 case study [Lopez-Lira & Tang \(2026\)](#): GPT-4 answers YES, NO or UNKNOWN to a fixed prompt about each headline; the score $s = 1, -1, 0$ drives a daily long-short portfolio whose drift earns 0.34% per day, Sharpe ratio 2.97, on overnight news (Table 1)
- ▶ Data: the FNSPID headlines of the 16 stocks of Lecture 15 [Dong, Fan & Peng \(2024\)](#); adjusted closes of the stocks and of SPY from data/market; day d = first trading day on or after the headline date, so $d + 2$ is the first return surely after publication
- ▶ Model: Qwen2.5-7B-Instruct [Qwen Team \(2024\)](#), open weights, temperature 0, with the prompt of Section 5 of the paper copied word for word (company name and headline inserted)
- ▶ Model problems: C1 (strategy, costs, Newey–West), B5 (clustered standard errors), B10 (wild cluster bootstrap), B7 (memory of the model)

C3: Replicating Lopez-Lira and Tang on FNSPID (task 2/3) [Proposed]

Solution discussed in class

Task

□ **Portfolio, calendar and cost conventions, fixed before any run**

- ▶ **Weights (Table 1):** long leg = equal-weighted mean return of the stocks with $s > 0$, short leg = same for $s < 0$, a leg only with at least two stocks; long-short = long minus short (gross exposure 2); if only one leg qualifies, hold it alone (footnote 20)
- ▶ **Returns:** raw stock returns, as in Table 1; the SPY-hedged version (each position minus SPY, as in C1) is a course adaptation, reported as a sensitivity: with both legs the hedge cancels, so it matters only on one-leg days
- ▶ **Calendar:** all trading days of the sample, as in C1, with return 0 on days without a position; mean, Newey–West t and Sharpe ratio over all days; hit rate over the days with a position
- ▶ **Costs (Figure 4):** c bp per round trip on each unit of notional; every position opens at the close of $d + 1$ and closes one day later, so the daily cost is c times the gross exposure ($2c$ with two legs, c with one), plus, in the hedged version, c on the net SPY notional (1 on one-leg days, 0 otherwise)
- ▶ **Regressions (d):** raw returns in % per day, as in Table 2; the date fixed effects absorb the market return; wild cluster bootstrap (B10) with the null model keeping both fixed effects: y^* = its fitted values plus its residuals times the stock weights w_g , then refit the full model; with dependence across stocks it is a sensitivity check

C3: Replicating Lopez-Lira and Tang on FNSPID (task 3/3) [Proposed]

Solution discussed in class

Task

What to do

- ▶ (a) Score every headline with the published prompt, parse only the first line and map YES to 1, UNKNOWN to 0 and NO to -1; report the proportion of responses in each category and the proportion with an invalid format
- ▶ (b) Average the scores by stock-day and form the daily portfolio with the conventions of the previous slide, earning the returns of day $d + 2$
- ▶ (c) Report the mean daily return in bp, its Newey–West t , the hit rate, the annualised Sharpe ratio and the shares of days with one leg and with no position
- ▶ (d) Estimate Eq. (1) at headline level (return of day $d + h$ on the score, stock and date fixed effects, $h = 0, 2$) with standard errors clustered by date and stock and the wild cluster bootstrap p -value over stocks (previous slide)
- ▶ (e) Recompute the Sharpe ratio net of 5, 10 and 20 bp per round trip (Figure 4); compare with C1's net-score strategy recomputed on the same stock-days at the same cost per unit of notional

What to report

- ▶ Your numbers next to Tables 1 and 2 of the paper, a chart of cumulative returns before and after costs, a Quantlet folder (code, Metainfo, charts)
- ▶ Interpretation: which of the differences from the paper (timing at $d + 2$ instead of the next open, 16 large stocks instead of 4,123, a smaller model, headlines older than the model) can your data separate, and what can B7 say about the bias of your estimate?

Common Pitfalls

- ▣ Reporting accuracy without the majority-class benchmark and macro-F1
- ▣ Evaluating a model on data that overlap its training data (FinBERT on PhraseBank)
- ▣ Comparing two classifiers through separate intervals instead of a paired test
- ▣ Matching a date-only headline to the same-day return and calling it a forecast
- ▣ Backtesting an LLM on news older than the release of its weights
- ▣ Choosing the prompt, the threshold or the model after seeing the test results
- ▣ Ignoring costs and the number of clusters: four trades a day, only 16 stocks
- ▣ Treating a model score as data: slopes are attenuated, and two model scores may share their errors

Take-Home Assignment (Team Project)

- Choose a text source not covered in class: BVB (Bucharest Stock Exchange) announcements, Romanian financial news, central-bank statements or crypto posts
- Label at least 200 texts yourselves (two annotators, report the agreement) for the sentiment about the asset mentioned; tune on one part and evaluate only on a held-out part (or out-of-fold)
- Compare LM, FinBERT, an open LLM and embeddings + logistic regression: accuracy and macro-F1 with CI, McNemar tests; for Romanian texts say how you apply the English LM and FinBERT (translation or a multilingual model)
- Build a signal with correct timing; report the period after the LLM's release separately
- Declare AI use in [AI_USE.md](#) and log every AI error you catch in [AI_ERRORS.md](#) (at least three per project, as in C2)
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

-  Natural Language Processing
-  The Next Word Problem
-  DELTA
-  LDA Extensions

Code and data

- Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_15

References (1/3)

- ▣ Aigner, D.J. (1973). Regression with a binary independent variable subject to errors of observation. *Journal of Econometrics*, 1(1), 49–59.
- ▣ Angelopoulos, A.N., Bates, S., Fannjiang, C., Jordan, M.I., Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671), 669–674.
- ▣ Araci, D. (2019). FinBERT: financial sentiment analysis with pre-trained language models. arXiv:1908.10063.
- ▣ Cameron, A.C., Gelbach, J.B., Miller, D.L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.
- ▣ Dawid, A.P., Skene, A.M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. *Applied Statistics*, 28(1), 20–28.
- ▣ DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.
- ▣ Dong, Z., Fan, X., Peng, Z. (2024). FNSPID: a comprehensive financial news dataset in time series. *Proceedings of the 30th ACM SIGKDD Conference*, 4918–4927.
- ▣ Fleiss, J.L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5), 378–382.

References (2/3)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Glasserman, P., Lin, C. (2023). *Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis*. arXiv:2309.17322.
- ▣ Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q. (2017). *On calibration of modern neural networks*. ICML; arXiv:1706.04599.
- ▣ Lopez-Lira, A., Tang, Y. (2026). *Can ChatGPT forecast stock price movements? Return predictability and large language models*. *Journal of Financial Economics*, 184, 104335.
- ▣ Loughran, T., McDonald, B. (2011). *When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks*. *Journal of Finance*, 66(1), 35–65.
- ▣ Malo, P., Sinha, A., Korhonen, P., Wallenius, J., Takala, P. (2014). *Good debt or bad debt: detecting semantic orientations in economic texts*. *Journal of the Association for Information Science and Technology*, 65(4), 782–796.
- ▣ McNemar, Q. (1947). *Note on the sampling error of the difference between correlated proportions or percentages*. *Psychometrika*, 12(2), 153–157.
- ▣ Newey, W.K., West, K.D. (1987). *A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix*. *Econometrica*, 55(3), 703–708.

References (3/3)

- ▣ Petersen, M.A. (2009). [Estimating standard errors in finance panel data sets: comparing approaches](#). *Review of Financial Studies*, 22(1), 435–480.
- ▣ Qwen Team (2024). [Qwen2.5 technical report](#). arXiv:2412.15115.
- ▣ Sarkar, S.K., Vafa, K. (2024). [Lookahead bias in pretrained language models](#). SSRN Working Paper 4754678.
- ▣ Shao, X. (2010). [The dependent wild bootstrap](#). *Journal of the American Statistical Association*, 105(489), 218–235.
- ▣ Simonsohn, U., Simmons, J.P., Nelson, L.D. (2020). [Specification curve analysis](#). *Nature Human Behaviour*, 4(11), 1208–1214.
- ▣ Tetlock, P.C. (2007). [Giving content to investor sentiment: the role of media in the stock market](#). *Journal of Finance*, 62(3), 1139–1168.
- ▣ Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M. (2020). [MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained Transformers](#). *NeurIPS*; arXiv:2002.10957.